KAI WICKE

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Second-year Mechanical Engineering student at Northeastern seeking a **Summer 2027 co-op or internship**, ideally where engineering meets AI. I build things where CAD, electronics and code meet: a finite element solver, computer-vision robots, and web tools I use every day.

EDUCATION

Northeastern University — Bachelor of Science, Mechanical Engineering 2025 – 2029

NU London (first year), Boston

Coursework: Statics, Materials Science with lab, Physics for Engineering, Calculus III, Differential Equations, Engineering Design

St. Mary's College High School, Berkeley, CA 2021 – 2025

3.88 GPA, 6 AP courses | Spirit and Athletic Director (elected) | Varsity Golf (4 years)

PROJECTS

Plane Stress FEA — 2D finite element solver 2026

- Wrote a finite element solver from scratch in TypeScript with no FEA or linear algebra libraries: meshing, 3- and 6-node triangle elements, a conjugate-gradient solver and stress recovery
- Validated against the Kirsch (plate with a hole) and Timoshenko (cantilever) analytical solutions

FocusFinder — attention-tracking desk robot 2026

- Raspberry Pi Zero 2W with OpenCV face detection; OLED eyes follow the user and a Pico sounds a buzzer when focus drops

Autonomous car 2025

- Built the chassis, wiring and Arduino C++ code; steers around walls and obstacles using three ultrasonic sensors

SwingIQ — golf swing analyser 2026

- MediaPipe pose tracking finds the phases of a swing and measures tempo, posture, sway and balance (React, TypeScript)

chess.ai — chess trainer 2026

- Analyses Lichess and Chess.com games with in-browser Stockfish and coaches play against bots rated 400–2400 (Next.js)

Life OS — personal dashboard and voice briefing 2026

- Dashboard of Gmail, Canvas, classes and tasks that ranks email by importance; Siri reads a daily briefing (Node.js, Vercel)

EXPERIENCE

Ole Factorie Candles, Oakland, CA Dec 2024 – May 2025, Jun – Aug 2026

Operations and Events (summer 2026); Retail Salesperson (2024–2025)

- Managed orders for the shop's largest clients, including Oakland International Airport (OAK)
- Planned and ran the shop's events
- Designed and built the company website

Freelance Web Designer, Boston, MA Nov 2026 – present

- Built a services page and contacted 30+ local businesses after auditing their websites for problems

Summer Camp Counselor, JCC East Bay, Oakland, CA May 2023 – Jul 2025

- Led activities and science experiments for groups of 10–20 campers aged 5–12, including children with disabilities

Math and Chemistry Tutor, self-employed, Oakland, CA Sep 2024 – May 2025

SKILLS

Engineering: SolidWorks, Fusion 360, 3D printing and prototyping, circuit design, Raspberry Pi, Arduino, finite element analysis
Programming: Python, C++, TypeScript and JavaScript, React, Next.js, Node.js, Git | **Computer vision:** OpenCV, MediaPipe
Interests: Golf, weightlifting, chess, hiking, mathematics